Finance Data Analysis

Project 5 — Multi-Domain Data Analysis Portfolio

Executive Summary

This project analyzes 1,500 simulated business-day financial observations from January 1, 2021 to October 1, 2026. The analysis covers price trends, daily returns, trading volume, moving averages, rolling volatility, monthly performance, correlations, and data-quality validation.

Important: The dataset is simulated for educational and portfolio purposes. Results are not real investment performance or financial advice.

Key Performance Indicators

Metric	Value
Starting Price	101.04
Ending Price	686.77
Overall Price Return	579.68%
Average Daily Return	0.16%
Daily Return Volatility	1.98%
Best Daily Return	7.76%
Worst Daily Return	-6.43%
Positive Return Days	53.13%
Negative Return Days	46.80%
Bullish MA Days	61.53%
Bearish MA Days	38.47%

Analytical Findings

Price Trend

The simulated closing price increased from 101.04 to 686.77, representing a 579.68% overall price return.

Monthly Performance

June had the highest average daily return among calendar months at 0.378%, while January had the lowest at -0.188%.

Monthly Extremes

October 2026 had the highest monthly average return at 1.24%, while March 2026 had the lowest at -0.83%. October 2026 is a partial month because the dataset ends on October 1.

Volatility

Maximum 20-day rolling volatility was 3.19% on June 23, 2025; minimum volatility was 1.10% on March 28, 2022.

Trading Volume

Highest trading volume was 999,751 on December 18, 2023, when the daily return was -1.07%.

Moving Averages

Bullish moving-average conditions occurred on 61.53% of trading days.

Correlation

Volume had very weak linear correlation with daily return (0.045) and volatility (-0.041). Daily return and 20-day volatility were also nearly uncorrelated (0.004).

Visualizations Produced

• Finance correlation matrix

• Stock price trend with 20-day and 50-day moving averages

• Daily returns over time

• Trading volume over time

• 20-day rolling volatility

• Average daily return by calendar month

• Moving-average signal distribution

Recommendations

• Monitor short- and medium-term price trends using moving averages alongside price data.

• Track rolling volatility to identify periods of elevated uncertainty.

• Evaluate both positive and negative return distributions rather than relying only on cumulative performance.

• Investigate unusually high trading-volume periods together with price and return movements.

• Use multiple indicators together instead of relying on a single metric.

• For real-world use, replace simulated data with validated market data and appropriate financial risk methodology.

Data Quality & Validation

• Final dataset: 1,500 rows × 16 columns.

• Duplicate rows: 0; unique dates: 1,500.

• Original missing values: 89, all from expected initial rolling-window calculations.

• Clean analysis dataset: 1,451 rows × 16 columns with 0 missing values.

• OHLC integrity checks passed.

• All prices and trading volumes are positive.

• Date range: January 1, 2021 to October 1, 2026.

Conclusion

The project demonstrates an end-to-end financial data-analysis workflow using Python, Pandas, NumPy, and Matplotlib, covering data preparation, validation, descriptive statistics, trend analysis, risk analysis, correlation analysis, visualization, and business-oriented recommendations.